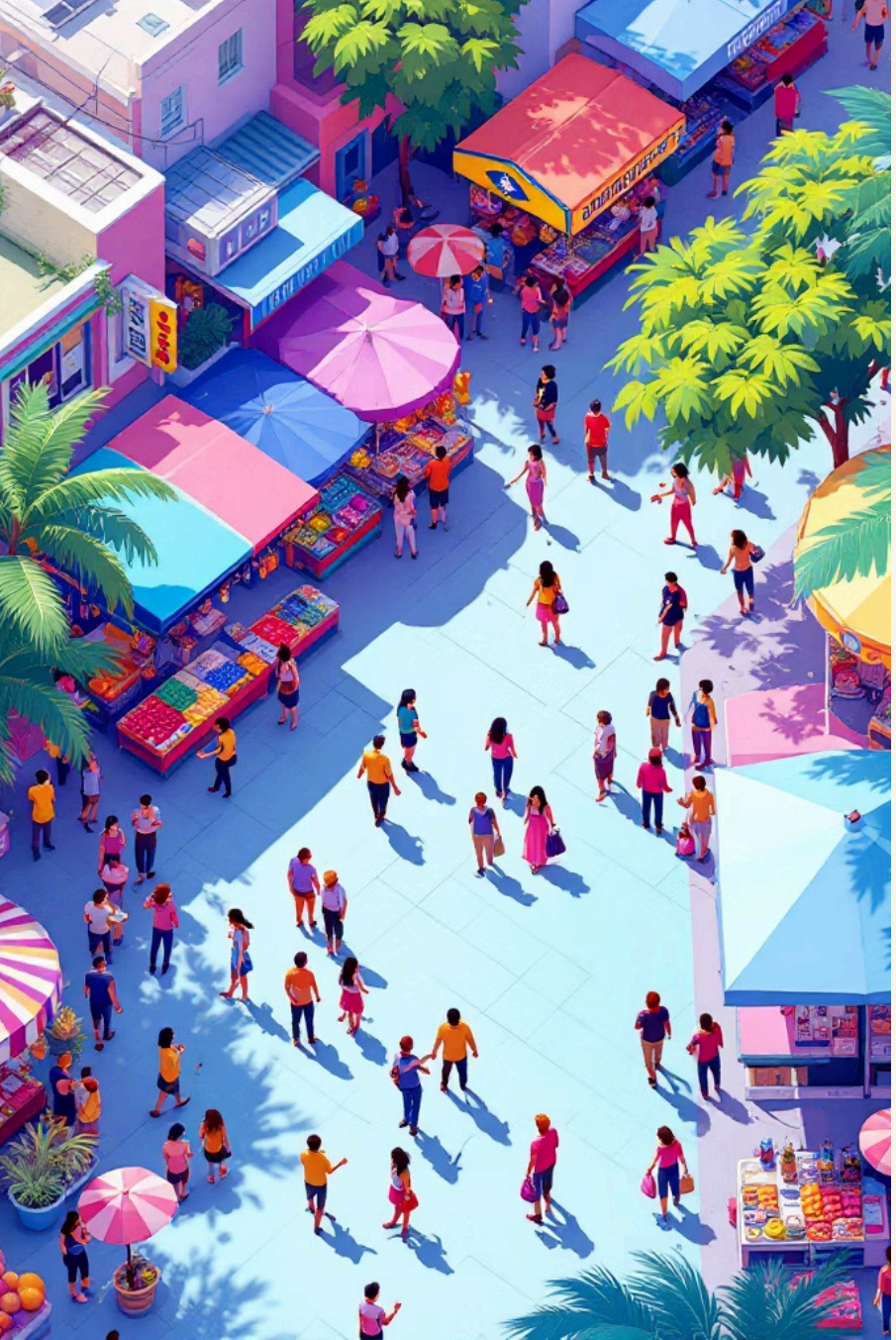

Customer Retention & Lifetime Value Analysis

Leveraging Customer Behavioural Analytics to Improve Retention and Revenue

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Kaggle Source: Brazilian E-Commerce Marketplace Dataset (Olist)

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The Retention Problem in Olist Marketplace

Business Context

Olist platforms invest heavily in customer acquisition — yet many customers make **only a single purchase** and never return, eroding the value of that investment (ie, CPA).

Three Core Challenges

- High customer **churn** with no early detection mechanism
- Low **repeat purchase rates** limiting revenue growth
- Limited visibility into **long-term customer value**

Project Objective

Identify high-value segments · Detect early churn signals · Deliver data-driven retention recommendations

Move from **mass marketing** to **targeted retention strategies grounded in customer behaviour**.

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Executive Summary

Olist should **shift from reactive churn management to proactive customer engagement strategies** using behavioural analytics

100K+ Orders Analysed

Customer engagement dynamics — not spending alone — strongly influence repeat purchasing behaviour

Early Churn Signals

Purchase rhythm and inactivity periods provide measurable early warning signals of customer disengagement.

Integrated Modelling

Combining RFM segmentation, CLV modelling, and churn prediction enables precise, targeted retention strategies.

Revenue Impact

Even modest retention improvements generate significant long-term revenue growth for this marketplace platform.

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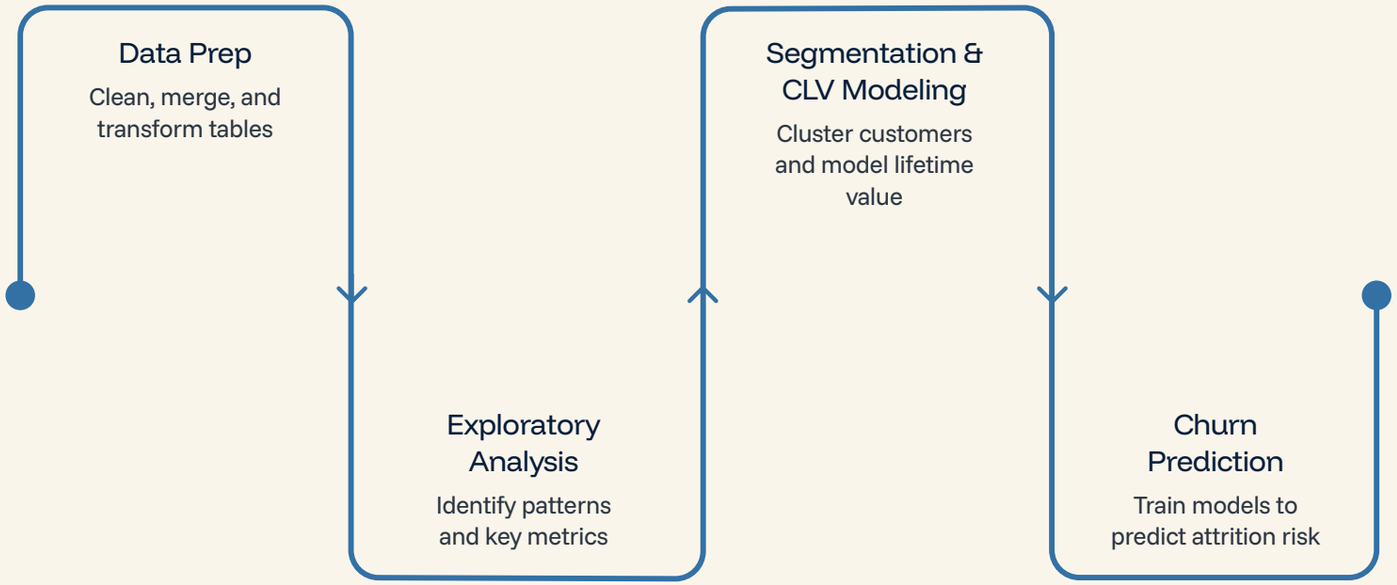
From Raw Transactions to a Structured Customer Analytics Dataset

Dataset Overview

Brazilian E-Commerce Dataset (Olist) — over 100,000 transactions spanning orders, customers, payments, and product information across a multi-table relational schema.

Data Preparation

- Merging multiple relational tables into a unified view
- Cleaning missing and inconsistent records
- Transforming timestamps for behavioural analysis
- Aggregating order-level data to the customer level



This structured pipeline ensures every analytical output is grounded in clean, customer-level behavioural data.

Establish **a centralised customer analytics dataset** to support continuous behavioural monitoring.

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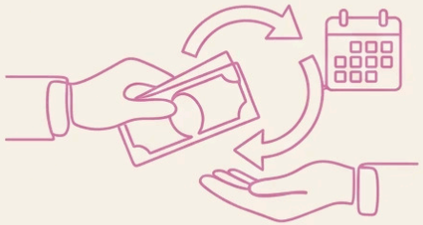
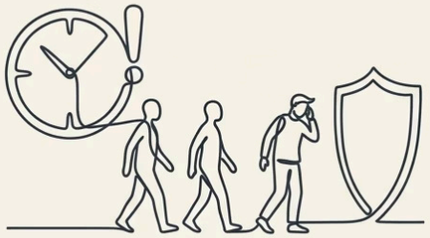
Segmentation & CLV Reveal Who Olist's Customers Really Are

High-Value Loyal Customers (High R, F, M)



Active Repeat Buyers (Mod R, High F)



At-Risk Customers (Low R, Mod F)

Low Engagement Customers (Low R, F, M)

RFM Segmentation

Customers were scored across three dimensions — **Recency**, **Frequency**, and **Monetary Value** — to surface meaningful differences in engagement and value.

Customer Lifetime Value (CLV)

CLV analysis estimates the **long-term financial contribution** of each customer segment. Two approaches were evaluated:

- **Historical CLV** — based on actual past spend
- **Predictive CLV** — forward-looking estimation from behavioural signals

Prioritise **marketing investment toward high-value segments** rather than treating all customers equally.

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Machine Learning Models Were Developed to Detect Churn Before It Happens

Churn Definition

Only customers with **multiple purchases** were included. Churn was defined as **no purchase within a 90-day window** — creating a precise, actionable binary target.

Modelling Approach

Logistic Regression

Baseline interpretable model

Random Forest

Nonlinear behavioural modelling

SMOTE

Class-weighted learning was applied to handle class imbalance

Key Engineered Features

Customer Lifetime Duration

Time between first and last purchase

Purchase Frequency & Intensity

Total number of completed orders and orders per unit of customer lifetime.

Average Order Value

Avg. revenue per order

Inter-purchases Time & Recency

Avg. time between consecutive purchases & time since the last purchase relative to cutoff time.

Review Ranking & Delayed Delivery

Number of reviews left and if deliveres were on time or delayed

The **evaluation framework used** (ROC-AUC, Precision-Recall AUC) **ensures that the model assessed not only on accuracy but also on their ability to correctly identify customers at risk of churn.**

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Customer Monitoring Dashbaord

An interactive Tableau dashboard was developed to enable ongoing monitoring of customer segments and churn risk. This makes the analysis accessible and actionable for all business users.

Business Value

By translating complex analysis into a visual dashboard, Olist can transition from static reporting to continuous customer insight generation. This dynamic tool directly supports:

- More informed marketing decisions
- Proactive customer retention strategies
- Data-driven customer segmentation

> Integrate this dashboard into marketing and CRM workflows to facilitate continuous monitoring of customer engagement and churn risk, empowering agile decision-making.

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Behavioural Engagement Is the Strongest Predictor of Retention

1

Consistent Purchase Rhythms

Customers with regular purchasing cadences are significantly more likely to remain active over time.

2

Widening Purchase Gaps

Increasing intervals between purchases are measurable **early warning signals** of impending churn.

3

Spending ≠ Loyalty

Customer spend alone is **not a reliable predictor** of long-term loyalty or retention.

4

Behavioural Metrics Win

Engagement-based features significantly improve churn **prediction accuracy** over spend-based models.

Monitor **behavioural engagement signals continuously** to identify at-risk customers early and **act before disengagement becomes permanent.**

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Strategic Recommendations & Business Impact

1

Implement Churn Monitoring

Build systems to detect customers showing **declining engagement patterns** in real time.

2

Focus on High-CLV Segments

Allocate marketing resources where they generate the **highest long-term return**.

3

Launch Re-engagement Campaigns

Use personalised promotions to **re-engage customers before churn occurs**.

4

Integrate Behavioural Analytics

Embed engagement metrics into **campaign design and retention strategy** decisions.

Business Impact

↑ Repeat Purchases

Higher purchase frequency from retained customers

↓ Wasted Spend

More efficient marketing budget allocation

↑ Customer LTV

Greater long-term value per customer relationship

Even **small improvements in retention** can drive substantial, compounding revenue growth across the customer base.

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Thank You

Julian Fernandes

Github: https://github.com/fern-1210/IronHack---Final-Project---CLV-RFM-Machine-Learning/tree/main

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